

EXPERIMENT 1

A) SETTING up of Linux Programming Environment

Linux programming can be performed either online or offline. For this experiment, a Linux environment is required.

B) Online Method

Online Linux environments such as **CoCalc** can be used for programming without installing Linux directly on the computer.

C) Offline Methods

The common methods of using Linux offline are:

1. Dual Boot:

Two operating systems, such as Windows and Linux, are installed on the same computer. The required operating system can be selected during startup.

2. Single Boot:

Only one operating system is installed on the computer.

3. Bootable Pendrive:

A bootable USB drive containing Linux can be used to run or install Linux.

4. Virtual Machine:

Linux can be installed and run as a virtual machine inside an existing operating system using virtualization software.

Virtual Machine Manager / Hypervisor

A hypervisor is software used to create and manage virtual machines.

Type-1 Hypervisor (Bare Metal):

It runs directly on the computer hardware without a host operating system.

Type-2 Hypervisor (Hosted):

It runs as an application on an existing operating system.

Examples: **Oracle VirtualBox** and **VMware**.

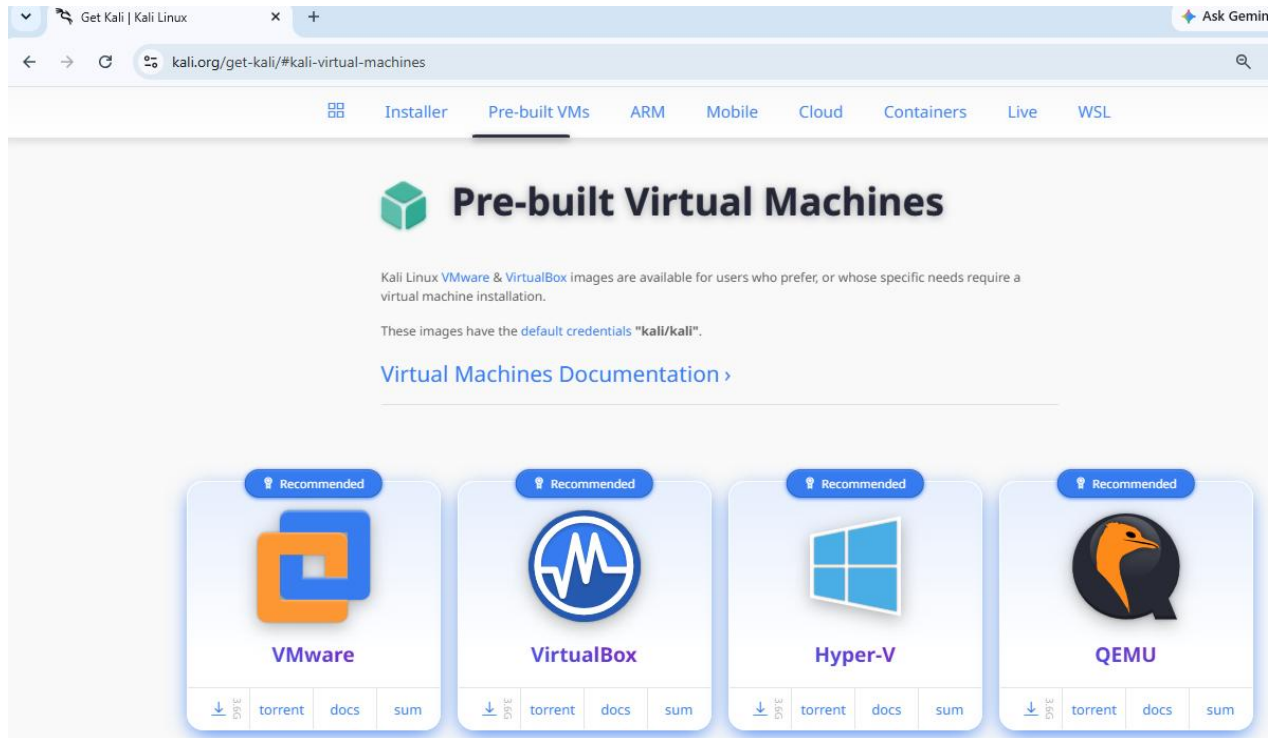
In this experiment, **Oracle VirtualBox** is used as the Type-2 Hypervisor and **Kali Linux** is used as the Virtual Machine.

Installation of VMM and VM

EXPERIMENT 1

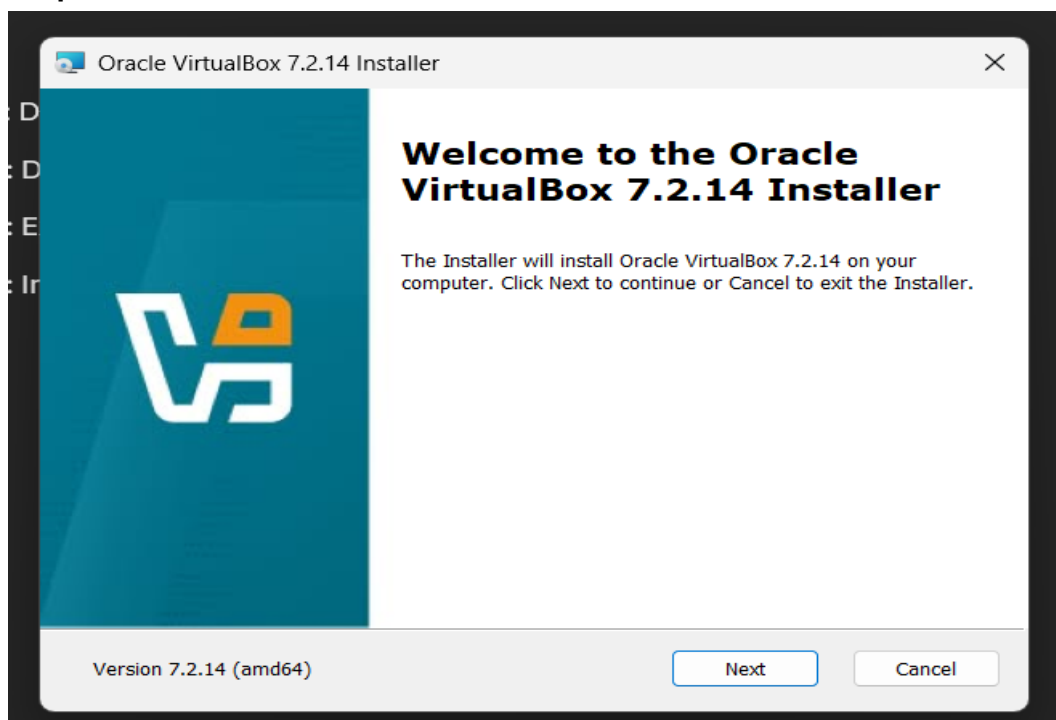
Step 1: Download Oracle VirtualBox for Windows hosts from the official VirtualBox website .

Step 2: Download the pre-built Kali Linux Virtual Machine for VirtualBox.



Step 3: Extract the downloaded Kali Linux file.

Step 4: Install Oracle VirtualBox.

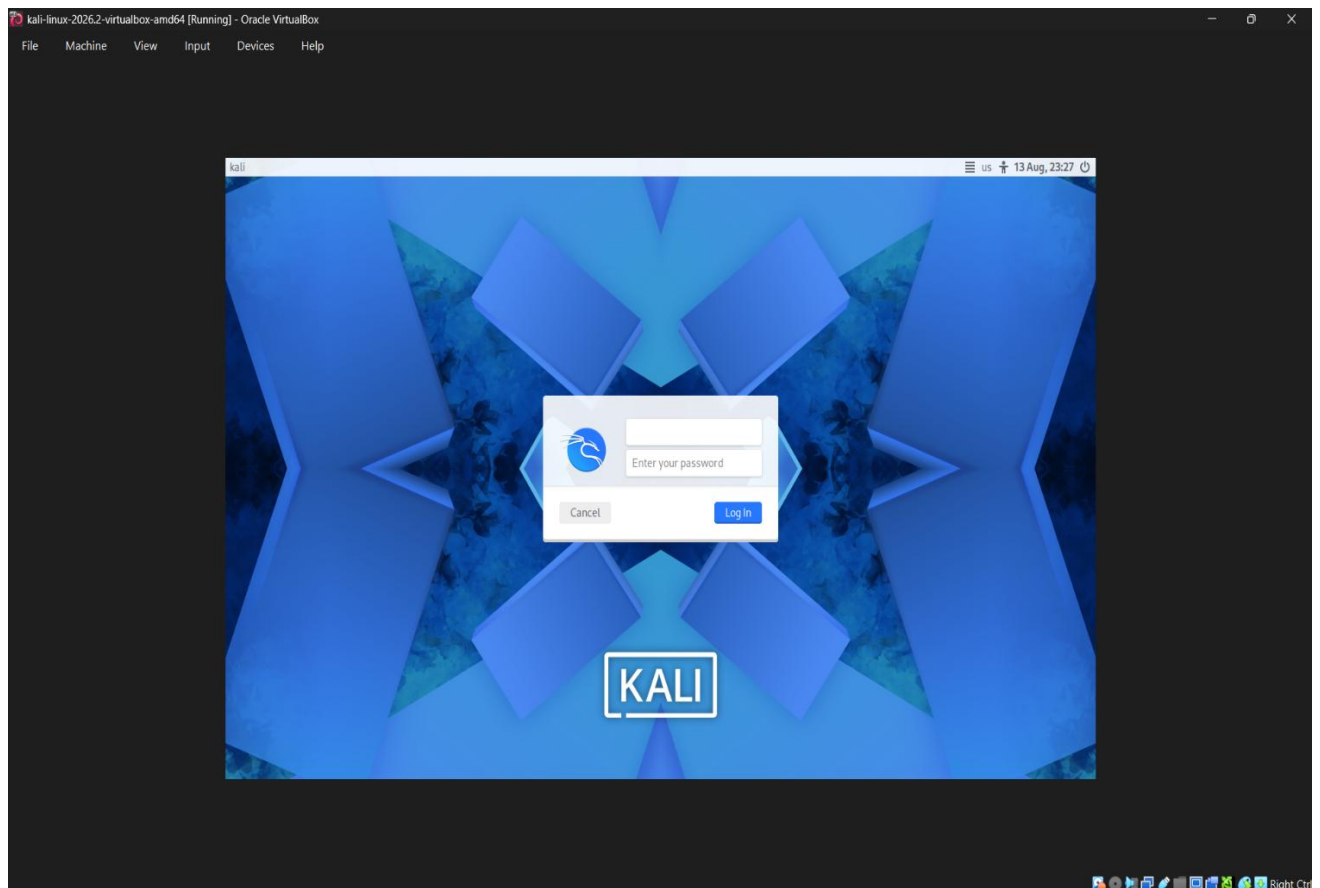


EXPERIMENT 1

Step 5: Open Oracle VirtualBox.

Step 6: Open/import the saved Kali Linux Virtual Machine.

Step 7: Start the Kali Linux Virtual Machine



B) Working with Linux Commands

The following Linux commands were executed in the Kali Linux terminal and their outputs were observed.

1. pwd

Command:

pwd

Description:

Displays the path of the present working directory.

Observed Output:

/home/kali

2. whoami

EXPERIMENT 1

Command:

whoami

Description:

Displays the name of the currently logged-in user.

Observed Output:

kali

3. ls

Command:

ls

Description:

Lists the files and directories present in the current directory.

Observed Output:

Folders such as Desktop, Documents, Downloads, Music, Pictures, Projects, Public, Templates and Videos were displayed.

4. mkdir

Command:

mkdir experiment1

Description:

Creates a new directory. In this experiment, a directory named experiment1 was created.

5. cd

Command:

cd experiment1

Description:

Changes the current working directory to the specified directory.

7. cd ..

Command:

cd ..

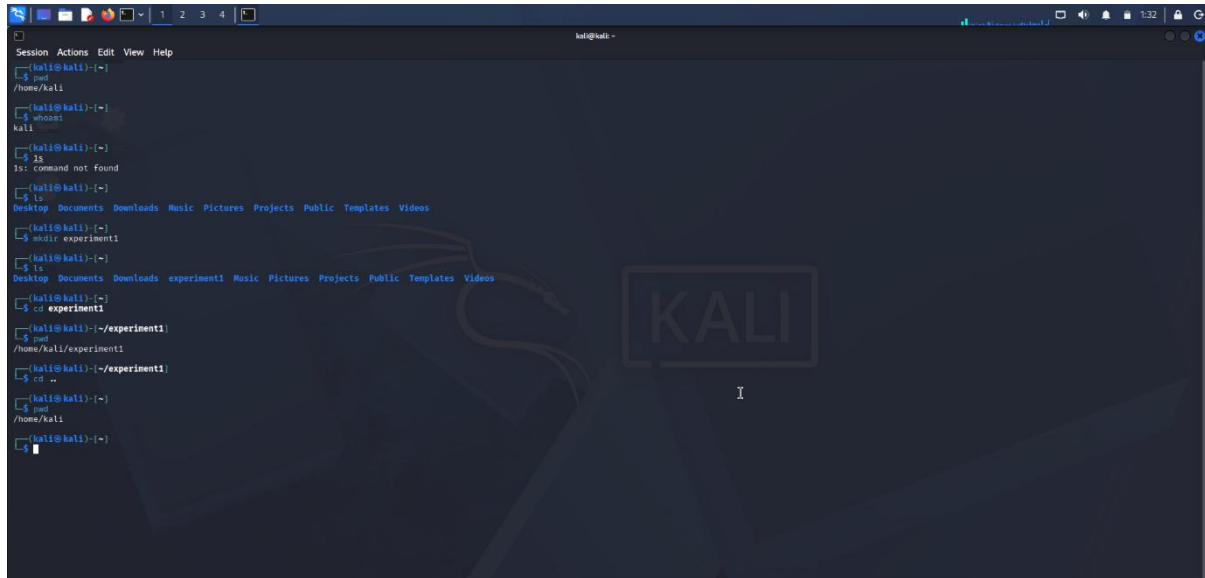
EXPERIMENT 1

Description:

Moves from the current directory to its parent directory.

Observed location after using pwd:

/home/kali

A screenshot of a Kali Linux terminal window. The terminal shows a series of commands and their outputs. The user starts in the root directory (~), then runs 'pwd' which outputs '/home/kali'. They then run 'whoami' which outputs 'kali'. Next, they run 'ls' which lists the contents of the home directory: Desktop, Documents, Downloads, Music, Pictures, Projects, Public, Templates, Videos. Then they run 'mkdir experiment1' to create a new directory. After that, they run 'ls' again, and the output now includes 'experiment1'. Finally, they run 'cd experiment1' to move into the new directory, and the prompt changes to '(kali@kali)-~/experiment1'. The terminal has a dark background with a 'KALI' logo watermark in the center.

8. rmdir

Command:

rmdir experiment1

Description:

Removes an empty directory. The experiment1 directory was removed using this command.

9. dir

Command:

dir

Description:

Displays the files and directories present in the current directory.

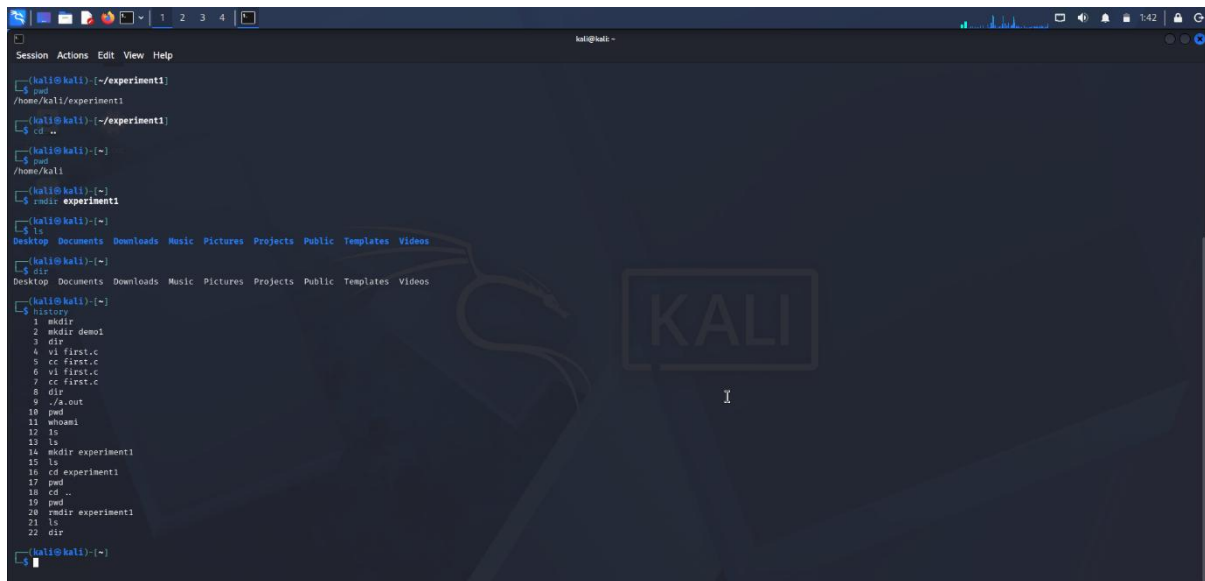
10. history Command:

history

Description:

Displays a numbered list of previously executed terminal commands

EXPERIMENT 1



```
(kali@kali): ~/experiment1
$ pwd
/home/kali/experiment1
(kali@kali): ~/experiment1
$ cd ..
(kali@kali): ~
$ pwd
/home/kali
(kali@kali): ~
$ mkdir experiment1
(kali@kali): ~
$ ls
Desktop Documents Downloads Music Pictures Projects Public Templates Videos
(kali@kali): ~
$ dir
Desktop Documents Downloads Music Pictures Projects Public Templates Videos
(kali@kali): ~
$ history
1 mkdir
2 mkdir demo1
3 dir
4 vi first.c
5 cc first.c
6 vi first.c
7 cc first.c
8 dir
9 ./A.out
10 pwd
11 whoami
12 ls
13 ls
14 mkdir experiment1
15 ls
16 cd experiment1
17 pwd
18 cd ..
19 pwd
20 mkdir experiment1
21 ls
22 dir
(kali@kali): ~
$
```

11. Clear Command:

clear

Description:

Clears the contents displayed on the terminal screen.

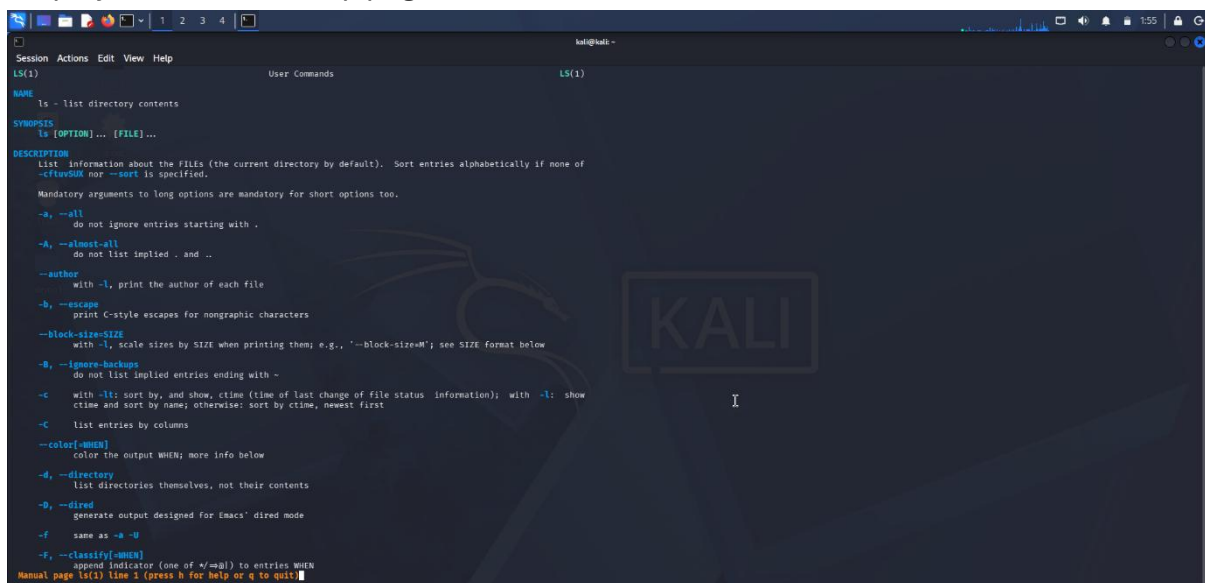
12. man

Command:

man ls

Description:

Displays the manual/help page for the ls command.



```
LS(1) User Commands LS(1)
NAME
ls - list directory contents
SYNOPSIS
ls [OPTION]... [FILE]...
DESCRIPTION
List information about the files (the current directory by default). Sort entries alphabetically if none of
--reverse nor --sort is specified.
Mandatory arguments to long options are mandatory for short options too.
-a, --all
do not ignore entries starting with .
-A, --almost-all
do not list implied . and ..
--author
with -l, print the author of each file
-b, --escape
print C-style escapes for nongraphic characters
--block-size=SIZE
with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below
-B, --ignore-backups
do not list implied entries ending with ~
-c
with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show
ctime and sort by name; otherwise: sort by ctime, newest first
-C
list entries by columns
--color[=WHEN]
color the output WHEN; more info below
-d, --directory
list directories themselves, not their contents
-D, --dired
generate output designed for Emacs' dired mode
-f
same as -a -U
-F, --classify[=WHEN]
append indicator (one of v/=a) to entries WHEN
Manual page ls(1) line 1 (press h for help or q to quit)
```

EXPERIMENT 1

File Commands

13. vi

Command:

vi sample.txt

Description:

Opens the vi text editor to create or edit a file.

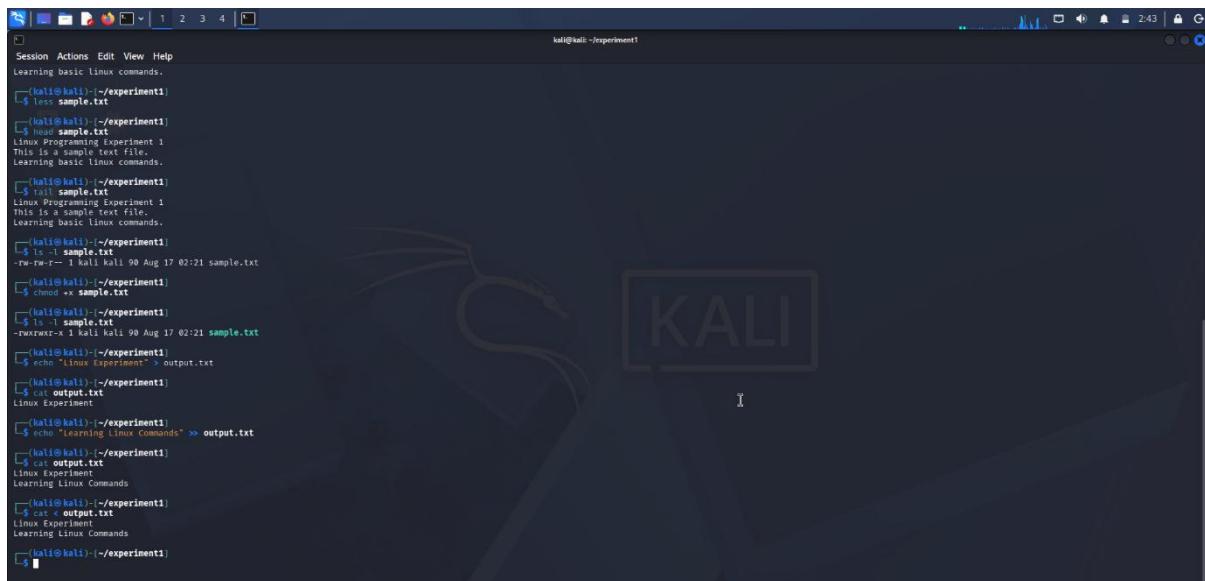
The following text was entered into the file:

Linux Programming Experiment

This is a sample text file.

Learning basic Linux commands.

The file was saved and closed using :wq.

A screenshot of a Kali Linux terminal window. The terminal shows a series of commands and their outputs. The commands include: less sample.txt, head sample.txt, tail sample.txt, ls -l sample.txt, chmod +x sample.txt, rm -rf sample.txt, echo "Linux Experiment" > output.txt, cat output.txt, and echo "Learning Linux Commands" >> output.txt. The outputs show the contents of the files and the results of the commands. The terminal has a dark background with a Kali Linux logo watermark. The window title is "kali@kali: ~/experiment1".

```
kali@kali:~/experiment1
$ less sample.txt
Linux Programming Experiment 1
This is a sample text file.
Learning basic Linux commands.

kali@kali:~/experiment1
$ head sample.txt
Linux Programming Experiment 1
This is a sample text file.
Learning basic Linux commands.

kali@kali:~/experiment1
$ tail sample.txt
Linux Programming Experiment 1
This is a sample text file.
Learning basic Linux commands.

kali@kali:~/experiment1
$ ls -l sample.txt
-rw-rw-r-- 1 kali kali 90 Aug 17 02:21 sample.txt

kali@kali:~/experiment1
$ chmod +x sample.txt

kali@kali:~/experiment1
$ rm -rf sample.txt
-rw-rw-r-- 1 kali kali 90 Aug 17 02:21 sample.txt

kali@kali:~/experiment1
$ echo "Linux Experiment" > output.txt

kali@kali:~/experiment1
$ cat output.txt
Linux Experiment

kali@kali:~/experiment1
$ echo "Learning Linux Commands" >> output.txt

kali@kali:~/experiment1
$ cat output.txt
Linux Experiment
Learning Linux Commands

kali@kali:~/experiment1
$
```

14. cat

Command:

cat sample.txt

Description:

Displays the contents of a text file on the terminal.

15. file

EXPERIMENT 1

Command:

file sample.txt

Description:

Displays information about the type of a file.

Observed Output:

sample.txt: ASCII text

16. cp**Command:**

cp sample.txt copy.txt

Description:

Copies a file. A copy of sample.txt was created as copy.txt.

17. mv**Command:**

mv copy.txt newfile.txt

Description:

Moves or renames a file. Here, copy.txt was renamed as newfile.txt.

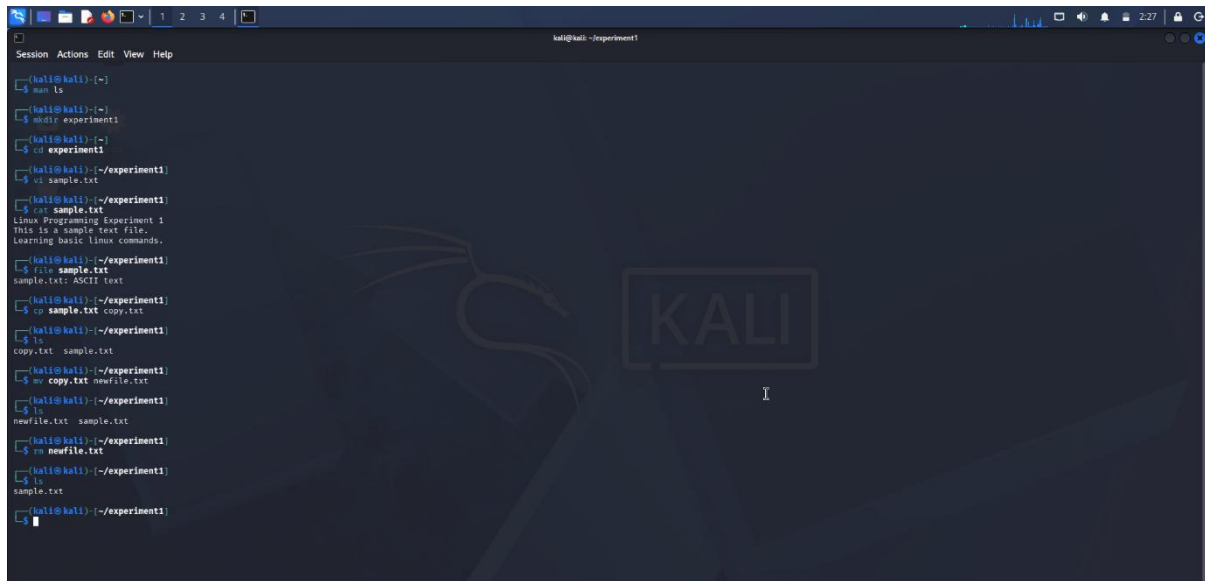
18. rm**Command:**

rm newfile.txt

EXPERIMENT 1

Description:

Removes a file from the directory.

A screenshot of a Kali Linux terminal window. The terminal shows a series of commands and their outputs. The user is in the directory ~/experiment1. The commands executed are: 'man ls', 'mkdir experiment1', 'cd experiment1', 'cat sample.txt', 'cat sample.txt', 'file sample.txt', 'cp sample.txt copy.txt', 'ls', 'mv copy.txt newfile.txt', 'ls', 'rm newfile.txt', and 'ls'. The output of 'cat sample.txt' shows the contents of the file: 'Linux Programming Experiment 1', 'This is a sample text file.', and 'Learning basic linux commands.' The terminal window has a dark background with a 'KALI' logo in the center. The top of the window shows the title bar with 'kali@kali: ~/experiment1' and standard window controls.

19. more

Command:

more sample.txt

Description:

Displays the contents of a file one screen at a time.

20. less

Command:

less sample.txt

Description:

Displays the contents of a file and allows navigation through the file.

21. head

Command:

head sample.txt

Description:

Displays the beginning lines of a file.

22. tail

EXPERIMENT 1

Command:

tail sample.txt

Description:

Displays the last lines of a file.

23. chmod

Commands:

ls -l sample.txt

chmod +x sample.txt

ls -l sample.txt

Description:

The chmod command is used to change file permissions. In this experiment, executable permission was added to sample.txt

Redirection Operators

24. > Output Redirection

Command:

echo "Linux Experiment" > output.txt

Description:

The > operator redirects output to a file. If the file already contains data, its previous contents are overwritten.

The command:

cat output.txt

displayed:

Linux Experiment

25. >> Append Redirection

Command:

echo "Learning Linux Commands" >> output.txt

EXPERIMENT 1

Description:

The >> operator appends new output to the end of an existing file without deleting its previous contents.

The command:

```
cat output.txt
```

displayed:

Linux Experiment

Learning Linux Commands

26. < Input Redirection

Command:

```
cat < output.txt
```

Description:

The < operator redirects the contents of a file as input to a command.

Observed Output:

Linux Experiment

Learning Linux Commands

```
kali@kali: ~/experiment1
Session Actions Edit View Help
Learning basic linux commands.
kali@kali:~/experiment1$ less sample.txt
kali@kali:~/experiment1$ head sample.txt
Linux Programming Experiment 1
This is a sample text file.
Learning basic linux commands.
kali@kali:~/experiment1$ cat sample.txt
Linux Programming Experiment 1
This is a sample text file.
Learning basic linux commands.
kali@kali:~/experiment1$ ls -l sample.txt
-rw-r--r-- 1 kali kali 90 Aug 17 02:21 sample.txt
kali@kali:~/experiment1$ chmod -x sample.txt
kali@kali:~/experiment1$ ls -l sample.txt
-rwxrwxr-x 1 kali kali 90 Aug 17 02:21 sample.txt
kali@kali:~/experiment1$ echo "Linux Experiment" > output.txt
kali@kali:~/experiment1$ cat output.txt
Linux Experiment
kali@kali:~/experiment1$ echo "Learning Linux Commands" >> output.txt
kali@kali:~/experiment1$ cat output.txt
Linux Experiment
Learning Linux Commands
kali@kali:~/experiment1$ cat < output.txt
Linux Experiment
Learning Linux Commands
kali@kali:~/experiment1$
```

EXPERIMENT 1

Result

The Linux programming environment using Oracle VirtualBox and Kali Linux was studied successfully. Basic Linux terminal commands, file manipulation commands, file permission commands and redirection operators were executed and their outputs were observed.

NAME - AYUSHI KUMAWAT

SAP ID - 590032252

COURSE - B.TECH CSE 2026-2030

BATCH - 16

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THANKYOU